



1. Simplify the the following configurations:

a) System 1:

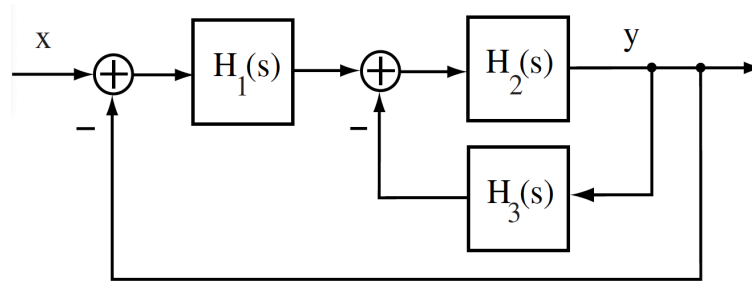


Figure 1:

b) System 2:

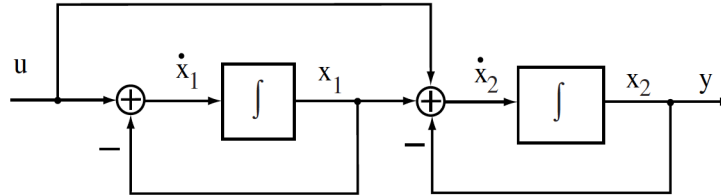


Figure 2:

2. A model of a brush DC motor without load is

$$L \frac{di}{dt} = v - Ri - K\omega$$

$$J \frac{d\omega}{dt} = Ki$$

where $R(\Omega)$ is the rotor resistance, $L(H)$ is the rotor inductance, $K(Nm/A)$ is the motor torque constant (also the back-emf constant), $J(kgm^2)$ is the inertia of the motor. The input of the system is the voltage $v(V)$ applied to the motor and the output is the rotor velocity $\omega(rad/s)$. The current $i(A)$ is considered to be an internal variable (or “state”)

a) Find the transfer function from v to ω . Give the values of the poles and zeros of the transfer function.

- b) Find the approximate transfer function that is obtained when $L = 0$, and give the values of its poles and zeros.
- c) Compare the numerical values obtained for parts (a) and (b) when $R = 0.7\Omega$, $L = 2.5 \times 10^{-3}H$, $K = 0.07Nm/A$, and $J = 5.7 \times 10^{-5}kgm^2$.
3. For each of the following transfer functions, write the corresponding differential equation
- a) $\frac{Y(s)}{U(s)} = \frac{7}{s^2+5s+3}$
- b) $\frac{Y(s)}{U(s)} = \frac{5}{(s+10)(s+11)}$
- c) $\frac{Y(s)}{U(s)} = \frac{s+3}{s^3+11s^2+12s+18}$
- d) $\frac{Y(s)}{U(s)} = \frac{s^5+2s^4+4s^3+s^2+4}{s^6+7s^5+3s^4+2s^3+s^2+5}$
- e) $\frac{Y(s)}{U(s)} = \frac{s^4+3s^3+2s^2+s+1}{s^5+4s^4+3s^3+2s^2+3s+2}$
4. Find the transfer function $\frac{I_2(s)}{V(s)}$ and $\frac{V_c(s)}{F(s)}$ for the following system:

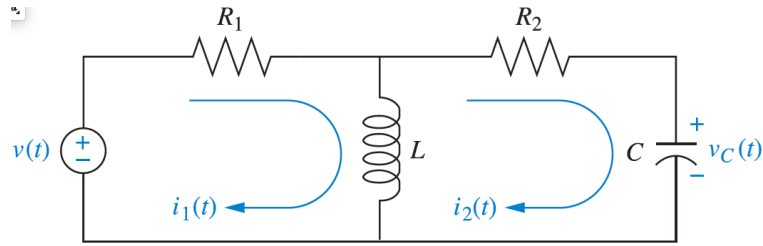


Figure 3:

5. Find the transfer function $\frac{X_2(s)}{F(s)}$ for the following system:

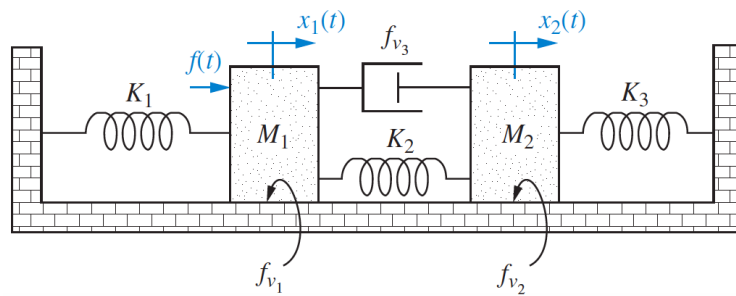


Figure 4: